Hints: Methods of proof – 2

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Lines: the n-th line meets the previous $n-1$ lines in $n-1$ distinct points. These points cut it into n pieces, and each piece splits one old region into two. So $f(n) = f(n-1) + n$ with $f(0) = 1$, which gives $f(n) = 1 + \frac{n(n+1)}{2} = \frac{n^2+n+2}{2}$.

Hint 2. In $\mathbb{R}^d$, let $f(n, d)$ be the number of regions cut out by n hyperplanes in general position. The n-th hyperplane $H \cong \mathbb{R}^{d-1}$ is cut by the others into $f(n-1, d-1)$ regions, because their traces on H are $n-1$ hyperplanes of H in general position. Each such region splits one old region.

Hint 3. So $f(n, d) = f(n-1, d) + f(n-1, d-1)$, with $f(0, d) = 1$ and $f(n, 0) = 1$. This is Pascal's rule, and the double induction on (n, d) (e.g. on $n + d$) gives $f(n, d) = \sum_{i=0}^{d} \binom{n}{i}$.

Problem 2

Hint 1. Let $\Delta P(x) = P(x+1) - P(x)$. If $\deg P = n$ and P is integer at $a, a+1, \ldots, a+n$, then ΔP has degree $n-1$ and is integer at $a, \ldots, a+n-1$. Induct on the degree: all $\Delta^k P(a)$, $k = 0, \ldots, n$, are integers, and $\Delta^{n+1} P = 0$.

Hint 2. Newton's forward-difference formula: $P(x) = \sum_{k=0}^{n} \Delta^k P(a) \binom{x-a}{k}$. Prove it by induction on n too, applying it to ΔP and using $\Delta\binom{x-a}{k} = \binom{x-a}{k-1}$. Hence P is integer at every integer.

Hint 3. The same argument with $a = 0$ shows that every integer-valued polynomial is an integer combination of the $\binom{x}{k}$. The $\binom{x}{k}$ are integer-valued and have distinct degrees, so they are linearly independent. Therefore they form a basis of the $\mathbb{Z}$-module of integer-valued polynomials.

Problem 3

Hint 1. Cauchy–Schwarz: the case of two terms, $(ab + cd)^2 \leqslant (a^2 + c^2)(b^2 + d^2)$, is equivalent to $(ad - bc)^2 \geqslant 0$. For the step, put $A = \left(\sum_{i \leqslant n} a_i^2\right)^{1/2}$ and $B = \left(\sum_{i \leqslant n} b_i^2\right)^{1/2}$. Then $\sum_{i \leqslant n+1} a_i b_i \leqslant AB + a_{n+1} b_{n+1}$ by the induction hypothesis. Apply the two-term case to (A, a_{n+1}) and (B, b_{n+1}).

Hint 2. AM–GM forward: the case $n = 2$ is $(\sqrt{x} - \sqrt{y})^2 \geqslant 0$. From n to $2n$, apply the case n to each half and then the case 2 to the two means. This gives all powers of 2.

Hint 3. Backward step $n \to n-1$: given $x_1, \ldots, x_{n-1}$ with mean m, apply the n-term inequality to $x_1, \ldots, x_{n-1}, m$. This gives $m^n \geqslant x_1 \cdots x_{n-1}\, m$, hence $m^{n-1} \geqslant x_1 \cdots x_{n-1}$.

Hint 4. Why the scheme: a direct step $n \to n+1$ is awkward, because the mean of $n+1$ numbers does not split into the mean of n of them plus one term in a way that AM–GM for n handles directly. Doubling and adding the mean as an extra variable are the natural moves. Together they reach every n, since every n lies below some power of 2.

Problem 4

Hint 1. Class equation: $|G| = |Z(G)| + \sum_i [G : C_G(x_i)]$, where the sum runs over representatives of the conjugacy classes with more than one element. Each such index divides p^k and is greater than 1, so it is divisible by p.

Hint 2. Hence $p \mid |Z(G)|$, and since $e \in Z(G)$, the centre is non-trivial. An induction on k is not really needed here. If you want one, use it for the corollary that a p-group has normal subgroups of every order: pass to G/Z, of order $p^{k'}$ with $k' < k$.

Hint 3. Order p^2: if $Z \neq G$, then $|Z| = p$ and G/Z has order p, so it is cyclic. If G/Z is cyclic, generated by gZ, then every element is $g^i z$ with $z \in Z$, and any two such elements commute. So G is abelian, which contradicts $Z \neq G$.

Problem 5

$(x^2 + y^2 + z^2 = 2xyz)$

Hint 1. Strengthen: show that $x^2 + y^2 + z^2 = 2^m xyz$ has no positive solutions for any $m \geqslant 1$. Look at the equation modulo 4, recalling that squares are 0 or 1 (mod 4).

Hint 2. The right side is even, so an even number of x, y, z are odd. If exactly two are odd, the left side is $\equiv 2 \pmod 4$, while the right side is divisible by 4 (it has a factor 2 and one even variable). So all three are even.

Hint 3. Substituting $x = 2x'$, $y = 2y'$, $z = 2z'$ gives $x'^2 + y'^2 + z'^2 = 2^{m+1} x'y'z'$, a solution of the same type with a smaller $x + y + z$. This is an infinite descent, so no solution exists.

$(x^3 + 2y^3 = 4z^3)$

Hint 1. Assume a non-trivial integer solution with $|x|+|y|+|z|$ minimal (the hint covers integers, not just positive ones). Since x^3 is even, x is even.

Hint 2. With $x = 2x_1$: $8x_1^3 + 2y^3 = 4z^3$, so $4x_1^3 + y^3 = 2z^3$, and y is even. With $y = 2y_1$: $2x_1^3 + 4y_1^3 = z^3$, and z is even. With $z = 2z_1$: $x_1^3 + 2y_1^3 = 4z_1^3$, a smaller solution.

Problem 6

Hint 1. Equivalence $A \sim PAQ$ with P, Q invertible means doing row and column operations. If $A \neq 0$, move a non-zero entry to position $(1,1)$, scale it to 1, and clear the rest of the first row and column.

Hint 2. This gives $\begin{pmatrix} 1 & 0 \\ 0 & A' \end{pmatrix}$ with $\operatorname{rank} A' = r - 1$. Apply the induction hypothesis on the rank to A'.

Hint 3. Nilpotent N ($N^k = 0$): N is not invertible, so there is a $v \neq 0$ with $Nv = 0$. In a basis starting with v, $N = \begin{pmatrix} 0 & * \\ 0 & N' \end{pmatrix}$, and N' is nilpotent because the block-triangular form gives $(N')^k$ as the lower-right block of $N^k = 0$. Make N' strictly upper triangular by induction on the size. Conjugating by the matrix $\operatorname{diag}(1, S)$ keeps the first column zero.

Problem 7

Note. "Cannot be proved by induction on any natural parameter" is not a precise mathematical statement. The hints read it as: no argument that works only with finitely many points or steps can prove it. Then they give the compactness proof and the rule.

Hint 1. Compare with sets where the statement fails: $f(x) = 1/x$ on $(0, 1]$, and $f(x) = 1/(x - 1/\sqrt{2})$ on $[0,1] \cap \mathbb{Q}$. Both are continuous, and every finite configuration of points behaves exactly as on $[0, 1]$. So any proof must use completeness of $[0, 1]$, which an induction over finitely many steps never sees.

Hint 2. Compactness proof: if f is unbounded, choose x_n with $|f(x_n)| > n$. By Bolzano–Weierstrass, a subsequence converges to some $x \in [0, 1]$, and continuity at x gives a contradiction.

Alternatively, let $s = \sup\{t : f \text{ is bounded on } [0,t]\}$. Show $s > 0$, that f is bounded on $[0,s]$, and that $s = 1$ ("real induction").

Hint 3. The rule: induction on $\mathbb{N}$ is available when the objects carry a natural-number parameter, and each object can be reduced to finitely many objects with smaller parameter. Behind this is well-foundedness. Over the reals, where descending sequences need not stop, it is replaced by sup, compactness, or a limit argument.

Problem 8

Hint 1. First prove Euler's formula $V - E + F = 2$ for connected plane graphs by induction on the number of edges. Removing an edge on a cycle merges two faces. If there is no cycle, the graph is a tree. Count edge–face incidences to get $E \leqslant 3V - 6$ for $V \geqslant 3$.

Hint 2. The sum of degrees is $2E \leqslant 6V - 12 < 6V$, so some vertex has degree at most 5. This property passes to subgraphs, which is what the induction on the number of vertices uses.

Hint 3. Five colours: remove a vertex v of degree $\leqslant 5$ and colour the rest by induction. If the neighbours use at most 4 colours, colour v. Otherwise v has five neighbours $v_1, \ldots, v_5$ in cyclic order, with five different colours.

Hint 4. Kempe chains: consider the subgraph spanned by colours 1 and 3. If v_1 and v_3 lie in different components of it, swap the two colours in the component of v_1, which frees colour 1 for v. Otherwise a 1–3 path joins v_1 to v_3. Together with v it forms a closed curve that separates v_2 from v_4, so the 2–4 swap works instead.

Problem 9

Hint 1. Maximal ideal: apply Zorn's lemma to the proper ideals, ordered by inclusion (the zero ideal shows the family is non-empty). The union of a chain of ideals is an ideal, and it is proper because it does not contain 1. This is exactly where the unit is used.

Hint 2. Maximal independent set: the same argument, as in the class work. The union of a chain of independent sets is independent, since linear dependence involves only finitely many vectors.

Hint 3. Countably generated $V = \operatorname{span}\{v_1, v_2, \ldots\}$: go through $v_1, v_2, \ldots$ and keep v_n if it is not in the span of the vectors kept so far. By induction the kept vectors are independent and span $\operatorname{span}\{v_1, \ldots, v_n\}$ at stage n. So they form a basis, and no choice principle beyond ordinary induction is needed.

Problem 10

Hint 1. Sperner in dimension n: take a triangulation of the simplex $[e_0, \ldots, e_n]$. Each vertex gets a label i from the coordinates that are positive at that vertex (i.e. from the indices of the smallest face containing it). Claim: the number of fully labelled small simplices is odd. Dimension 1: along the segment, the label changes from 0 to 1 an odd number of times.

Hint 2. Induction step (door counting): call an $(n-1)$-face labelled $0, \ldots, n-1$ a door. Count pairs (small simplex, door on its boundary). A fully labelled simplex has exactly one door, any other simplex has 0 or 2, and an interior door is counted twice.

Hint 3. The doors on the boundary all lie on the face $[e_0, \ldots, e_{n-1}]$, and by the induction hypothesis there are an odd number of them. Hence the number of fully labelled simplices is odd, in particular non-zero.

Hint 4. Brouwer: for continuous $f : \Delta \to \Delta$, label a vertex v by some i with $v_i > 0$ and $f(v)_i \leqslant v_i$. Such an i exists, since the coordinates of both v and $f(v)$ sum to 1. The labelling is a Sperner labelling. Take finer and finer triangulations, and use compactness to get a limit point x

of fully labelled simplices. Then $f(x)_i \leqslant x_i$ for all i, so $f(x) = x$. For $n = 1$ this is the intermediate value theorem applied to $f(x) - x$.